Do Fans Follow Last Seasons Record? Winning and NFL Home Attendance (2010-2019)

Vir Juneja · ECON 3916 Midterm Project · November 9th, 2026

Repository: *https://github.com/Vir-Juneja/ECON3916-MIDTERM* **Notebook:** *https://colab.research.google.com/drive/16-QCdPIStLJNiRPyhq6AbpIy8DnJcCfm?usp=sharing.*

Question	*For NFL teams from 2010 to 2019, how is a teams winning percentage in a season, associated with its average home attendance per game, holding last seasons winning percentage, the size of its metro area, the team, and the year fixed?*
Headline finding	*Within a team, a 0.1 rise in this seasons win % (1.6 more wins) goes with about 137 more fans per home game. The 95% interval for the full coefficient, -968 to 3706, includes zero. Last season's record seems to matter more, as a 0.1 rise is associated with about 458 more fans per game. The 95% interval was from 1666 to 7484. N = 320 team-seasons. They are both conditional associations, not causal effects.*
Data	*Pro Football Reference weekly attendance tables (2010-2019):* *Regular season game results from the nfl verse games file (2009-2019)* *Census Bureau metro-area population estimates (2010-2019)* *One row is one team-season, the final N is 320, with no missing values*
Method	*OLS of attendance per game on this season's win %, last seasons win %, log metro population, team dummies and year dummies. Standard errors were clustered by franchise, as a Breusch Pagan test rejected constant variance, and each franchise appears 10 times.*

Section 1

This project asks whether NFL fans come out in bigger numbers when their team wins. Economically, it asks how demand for a product responds to the quality of that product. A team's record is the most objective way to measure the quality of the product on the field, and home attendance is the amount of tickets fans buy. The answer matters to NFL teams, which use this data to set ticket prices, and decide how much money they want to spend in the off-season. A team consistently at capacity might want to look to renovate, and will have to evaluate whether or not their attendance is sustainable before investing. Additionally, many stadiums are built with tax-payer or state money, so it matters for cities. The amount of money is usually impacted by revenue of the team, and attendance (sold tickets), is a large driver of that, and the case for the money.

Attendance: This comes from Pro Football Reference, who post the attendance each teams reports for every game. I exported the data off their website for each season from 2010 to 2019, and then combined it into a larger excel sheet, using one sheet per season. The numbers are dependent on team decisions. Teams will report their attendance on tickets sold, and not the people actually in the game. Additionally, teams play abroad occasionally, and it counts as a home game for the team listed as the home team. For example when the Broncos and 49ers played in London, the Broncos were slated as the

home team and their attendance is tracked as such. Every team plays 8 home games per season, so to get attendance per game, it is the total home attendance/8.

Winning. Win % came from the nflverse games file, which is a record of every regular season score, along with teams total wins, and losses. This was cross referenced with Pro Football References data to ensure validity. I count each win as 1, a tie as .500, and a loss as 0, (per NFL standard practice) and average it over the teams games. The data starts from 2009, as we need the previous season record for 2010.

Market Size: Metro area population data came from the Census Bureaus yearly estimates for each metropolitan area. I went by hand and matched each team to their metro area. Some teams share a metro area, like the Giants and Jets(New York), the Raiders and 49ers (San-Francisco/Oakland), and the Rams and Chargers (LA). The Rams and Chargers both moved from different cities, so we treat the old team-city pair as its own team, though last seasons win % follows because it is still the same franchise.

What is missing: The final dataset had no missing values. All 320 team-seasons had every variable. There is a gap however, inside of the outcome. When a team sells out of tickets, the attendance = stadium capacity. Demand beyond that cannot be tracked, and those values become hidden because of what they would have been. The proper mechanism for this is MNAR, not MAR because there is nothing else that would have let me predict this hidden demand. This does matter for the answer, if a winning teams stadium is already full, the extra demand cannot be tracked in the data, which pushes the estimated impact of winning towards zero. Reported attendance also can move for reasons not related to fans, as teams can change how they count tickets.

Table 1. Summary statistics

Variable (units)	Mean	Median	SD	Min	Max	N
Home attendance per game (thousands of fans)	67.81	68.46	8.70	25.34	92.72	320
Win % this season (0 to 1)	.50	.50	0.19	0.00	0.94	320
Win % last season (0 to 1)	.50	.50	0.19	0.00	0.94	320
Metro population(millions)	4.87	3.50	4.44	0.31	19.33	320

Attendance from Pro Football Reference, 2010-2019. Win % from nflverse game results, 2009-2019.

Metro Population from Census Bureau estimates, 2010-2019. Each row is one team-season.

The average home game drew about 67,800 fans. The range was from 25,300 for the 2017 chargers, who had to play in a smaller temporary stadium while moving, to about 92,700 for the 2017 Cowboys. The Win % runs from 0 to .94. The average is .50, as when one team wins, another must lose. Metro population is skewed because New York and Los Angeles are much bigger than everywhere else, which is why we enter the model in logs.

Section 2

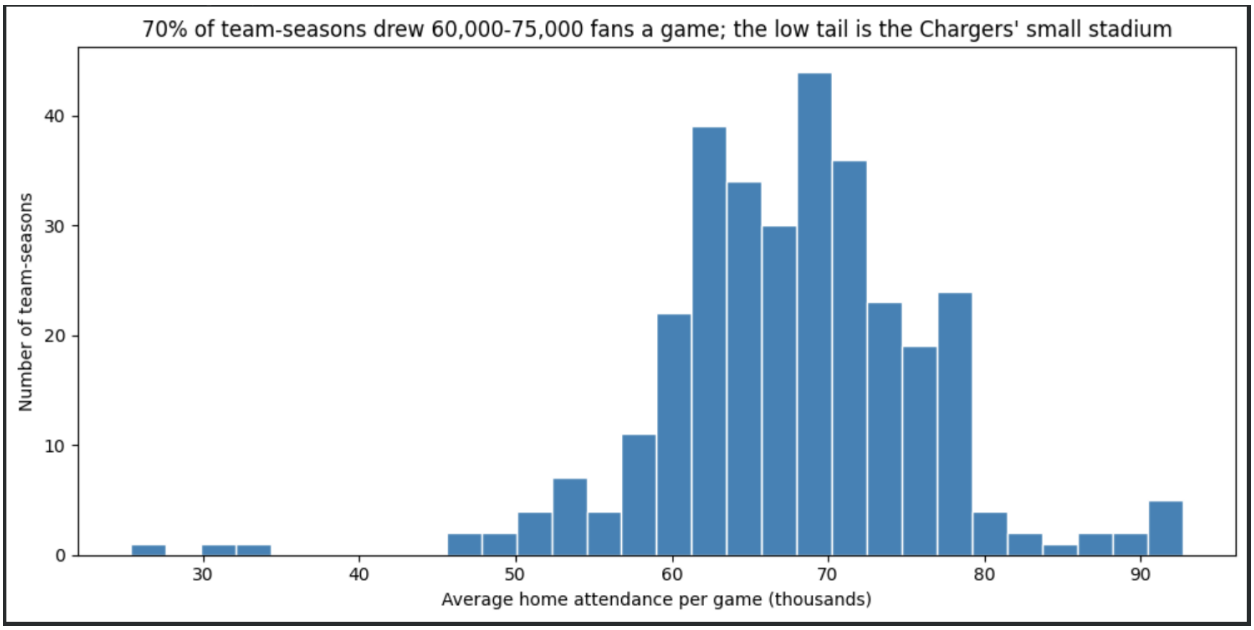


Figure 1. Seven in ten team-seasons drew between 60,000 and 75,000 fans per home game. The very low values are from the Chargers three seasons in a small, temporary stadium, so it gives attendance a short left tail instead of a long right one. Source: Pro Football Reference, 2010-2019

Figure 1 shows the outcome before any model. Most team-seasons sit in a band between 60,000 and 75,000 fans per game. Attendance is not right skewed in the way that income or prices are, it has a skew of -0.49, which is a slight left skew and caused by the Chargers small stadium. Its median (68,460) is close to its mean (67,806). I did not use a log as it does not have a long right tail, so I kept attendance in fans per game.

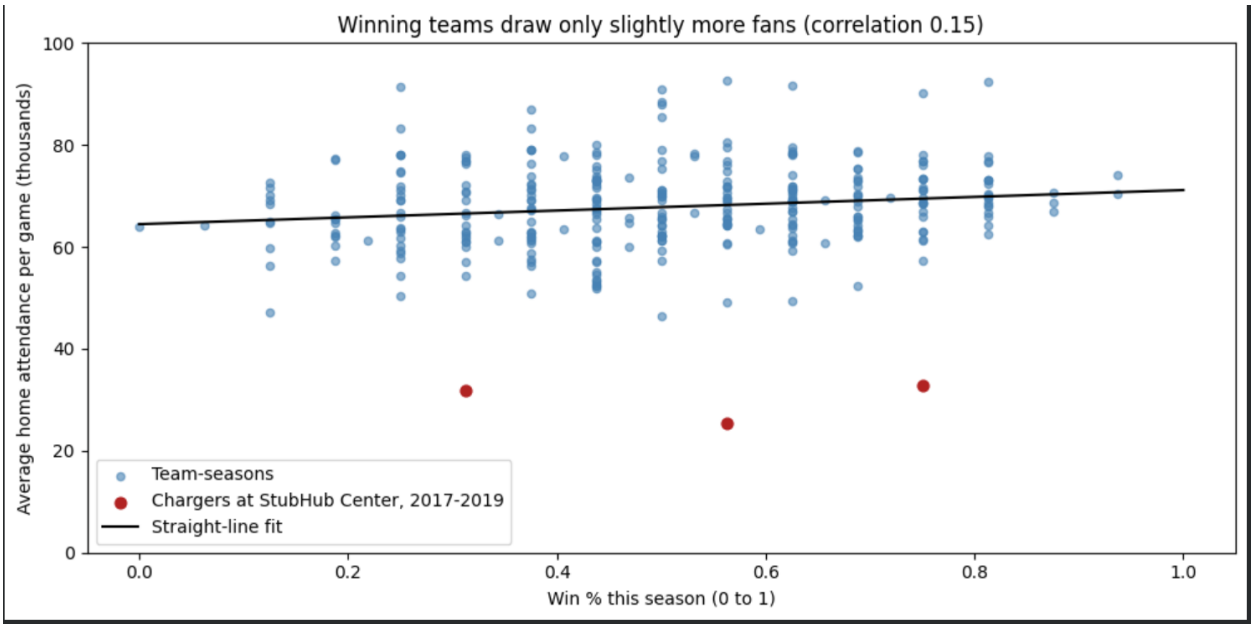


Figure 2. Winning teams only draw slightly more fans. The correlation is 0.15, the fitted line rises by about 6700 fans per game, from a winless season to an unbeaten one. Source : Pro Football Reference, nflverse.

Figure 2 plots attendance against the seasons win %. Teams that win end up drawing a few more fans, but not by a large amount. The correlation is only 0.15, so it is clearly not a strong linear relationship. The gap between 0 and 16 wins only leads to a 6700 increase in fans per game, which is about a tenth of an average crowd. The spread between teams at the same win % is much larger than that difference. The vertical axis starts at zero, so the slope is not made to look any steeper than it actually it is. This picture is a mix of two things, some teams always draw more fans, due to market, or their stadium, and those teams might also tend to win more. The model in Section 3 separates these two by comparing each team only with itself.

3. The Model

I fit the following model by OLS, where i is a team and t is a season:

Attendance per game(i,t) = b0 + b1 * Win%(I,t) + b2 * Win %(i,t-1) + b3 * log Metro population(i,t) + team dummy(i) + year dummy(t) + error(i,t).

The team dummies give every team their own baseline, so b1 is estimated from how a teams attendance moves when it wins more or less than usual for that team. The year dummies take out anything that moved attendance across the whole league in a given season. Column 1 of Table A1 in the appendix is the main model. Win % runs from 0 to 1. I describe a change of 0.1, in a 16 game season this is only 1.6 more wins

Win % this season (b1 = 1369). Holding last seasons record, metro size, and the team and the year fixed, a 0.1 rise in this season's win % goes with about 137 more fans per home game. This is a very small amount, about 0.2% of an average crowd of 67,806. The direction is positive, however, the estimate is not precise. The classical standard error (998) is close to the coefficient size. The 95% confidence interval for the full coefficient runs from about -597 to 3,334, and p = 0.17. I can't rule out that winning more this season makes no difference to a team's crowds at all. Section 4 shows that the clustered errors actually make this interval wider, and not narrower.

Win % last season(b2 = 4575). Holding this seasons record and everything else fixed, a 0.1 rise in last seasons win % goes with about 458 more fans per home game, which is about 0.7% of an average crowd. This estimate is precise, the 95% interval for the full coefficient runs from 2617 to 6533, which is well above zero, and $p < 0.001$. It seems that fans buy tickets based off of last years results, which fits how tickets are sold. Most season tickets are bought before the season starts, and the best information a fan has is the teams record from last year.

Why each control is there. Each control either blocks a confounding path, or helps to add precision. None of them are caused by this seasons win percentage.

Last seasons win % blocks a confounding path. It drives this years season ticket sales, and is a good predictor of record of this year, as most good teams will typically stay good. Without it, this seasons wins would be assigned credit for demand/sales that was locked in prior to the season. This would be a bad control if this seasons results could change it, but this is not possible as last years results are fixed and happened first.

Team dummies block confounding from things about each teams that remain the same every year. Fan loyalty, stadium size (barring a relocation), and the market area they are in. Column 3 of Table A1 drops the team and year dummies, which causes the coefficient on this seasons win % to triple to 4,468, and causes R squared to fall from .907 to .060. Most of the pattern in Figure 2 is good teams in big, loyal markets, and not teams drawing more interest when they win. The dummies would only be bad controls if winning could change what team you are, but it can not. Even when a team's name is changed, they retain their players, and are still the same team.

Year dummies help to add precision by absorbing changes that will affect every team in a season. This could be anything like a new TV deal, or a major change in the state of the economy. This would only be a bad control if one teams winning could change something league wide in a season, which is highly unlikely.

Log metro population is also there for precision. It only changes very slightly over the course of ten years, so the team dummies already capture most of it. It's estimate (-559) is close to zero with a huge standard error (8071), and dropping it does almost nothing to the win % coefficients. It would be a bad control if winning could drive city growth, which is very unlikely to happen.

I left out two variables on purpose. Since making the playoffs is caused by winning, it sits on the path from win % to attendance, and controlling for it would remove part of the effect that I am trying to measure. Ticket price is set with expected demand in mind, but I decided to treat it as a threat, rather than a control.

4. Diagnostics

Residual plots. Figure A1 in the appendix plots the main models residuals against both the fitted values and against this seasons win %. Since Neither plot shows a curve, a straight line in win% seems like a fair fit. The spread is not even, it is widest for team seasons with fitted attendance in the mid 50,000s to low 60,000s, and tightest for the highest drawing teams. This makes sense, as there is a capacity problem. Teams that already fill their stadium cannot sell more tickets, so their errors will bunch up in that area.

Constant Variance. The Breusch Pagan test rejects constant variance clearly(LM = 106.0, p = 7.9 * 10^-7), so the classical standard errors are too small. I used HC3 robust standard errors instead. Each franchise also appears ten times, and a teams good years and bad years for attendance usually come in runs. This made me believe that their errors are probably correlated over time, so I decided to cluster the standard errors by franchise (32 clusters). I treated the clustered errors as final.

With clustered errors, none of the conclusions change. The seasons win % is still imprecise, and its standard error rises from 998 to 1,146, p = 0.24, and the 95% interval runs from -968 to 3,706. Last seasons win % is also still very clearly positive, its standard error rises from 995 to 1426, p = 0.003, and the new interval runs from 1666 to 7484. The standard error grew by almost half when I clustered, which says that the repeated observations of each team mattered.

Variance Inflation Factors: Both win % variables have VIFs of about 1.4. That is low, which means although this seasons and last seasons records are related, neither is even close to being a copy of the other. Log metro population has a VIF of about 1699, and with team dummies in the model, it is almost a copy of them, which makes sense a cities population does not change much in 10 years. That explains the large standard error. As we have seen earlier, collinearity only affects the standard error of the variables involved, which is why dropping population did not change the win% results.

Influential Observations. Table A3 lists the five team seasons with the largest Cook's distance. 21 of the 320 are above the usual 4/n cutoff. Almost all are related to stadium issues. The Rams in 2016 and 2017

were their first two seasons back in Los Angeles, which drove huge crowds that dipped after the novelty faded. Washington in 2018 saw a huge drop due to concerns about ownership, and fans protested in an effort to force the owner to sell the team. The Vikings in 2014 and 2015 played in a smaller, temporary stadium.

The most influential was the Rams in 2016, with a Cook's distance of 0.156. Because of the team dummies, the model only compares the Rams against themselves. Their 2016 team had a losing record, but drew by far their biggest crowds, due to the novelty of their team. The model sees that as winning and attendance moving in opposite directions, which pulls the win % estimate down. When I drop it, the coefficient on this seasons win % rises from 1369 to 1939, which is about 0.57 of its standard error. It does not change either conclusion, this seasons win % estimate is still small and imprecise, and last seasons is positive, which shows the results are not contingent on one odd year.

5. Threats and Conclusion

For the coefficient on this seasons win % to be causal, nothing else that changes from year to year could affect both its winning and its crowds. The team and year dummies remove what is shared within a season, and what is fixed about a team, but threats still remain.

Price. Teams may raise ticket prices after winning, and higher prices drives out some fan demand. The model holds team fixed, but not price, so part of what winning does for demand is offset by the increase in price, which makes winning look like it matters less than it does.

Capacity. Sold out stadiums cap attendance, so demand above capacity can not be tracked, which also pushes the estimate towards zero.

Excitement: A new player, a new coach, or a returning player from injury could raise both expected wins and ticket sales at the same time. This would push the estimate up, and we saw this occur with excitement about the Rams move.

Reverse causation. It is possible that big, loud, crowds actually drive winning, so part of the arrow could run from attendance to winning, and not the other way.

Because these are real threats, and move things in different directions, my estimate is not a causal effect. It is a conditional association, within the same team, relative to the league in the same season, how home attendance moves with this seasons win %, holding both last seasons record and metro size fixed. This also applies to last seasons win %. A design like comparing teams both before and after a coaching change would need a clear comparison group, which cannot be checked.

Conclusion. Within a team, crowds move very little with this seasons winning. A 0.1 rise in win % goes with about 137 more fans per game, and zero is within the interval. Last seasons record is more important, a 0.1 rise is associated with about 458 more fans per game. and that result holds with robust and clustered standard errors, and after dropping the most influential season. A reader can be fairly confident that there is some link between last seasons record and this seasons attendance, but should not read this as the effect of winning on demand, because price, capacity, and excitement move with winning. With another month, I would add ticket prices to the model, use game to game attendance, and use each stadiums capacity to track sold out games.

Appendix

Table A1. Home attendance per game regressed on win% and controls.

	(1) Main	(2)No Population	**(3)No dummies**
Win % this season(0-1)	1369	1365	4468
	(998)	(995)	(2622)
Win % last season (0-1)	4575	4572	7770
	(995)	(992)	(2615)
Log Metro population	-559		1212
	(8071)		(582)
Team Dummies(34 team city units)	Yes	Yes	No
Year Dummies(2010-2019)	Yes	Yes	No
N	320	320	320
R^2	.907	.907	.060

Dependent variable : home attendance per game, in fans

Standard errors in parentheses: classical. Column 2 drops log metro population, which the team dummies nearly duplicate. Column 3 drops the team and year dummies. The constant and dummy coefficients are not shown.

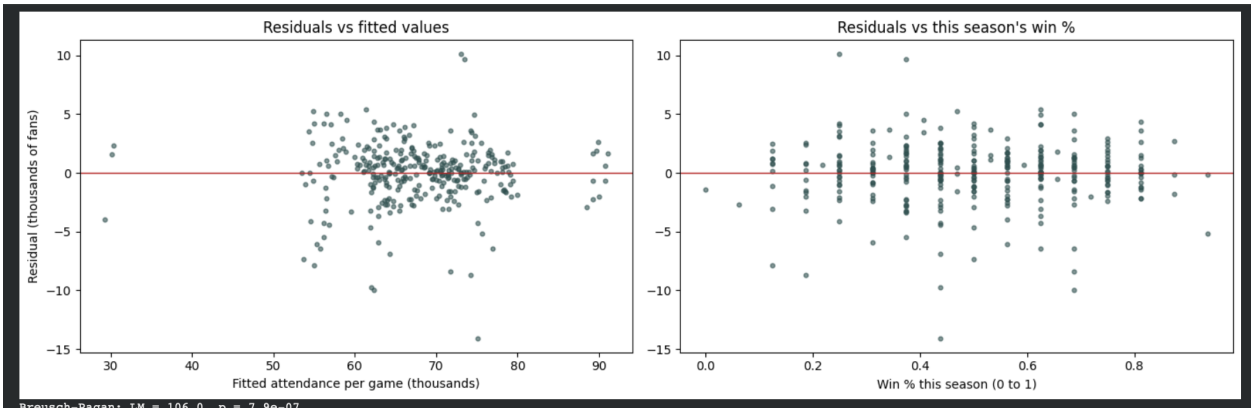


Figure A1. The main models residuals show no curve, but their spread is uneven, widest for teams with attendance in the mid 50,000s to low 60,000s, tightest for teams near 90,000.

Table A2: Classical, Robust, and clustered SE for the main model

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term                      coef  classical SE   HC3 SE  cluster SE   HC3 p  cluster p
-------------------------------------------------------------------------------------
Win % this season        1,369           998    1,145       1,146   0.233      0.241
Win % last season        4,575           995    1,122       1,426   0.000      0.003
Log metro population      -559         8,071    6,574      10,080   0.932      0.956
Clusters: 32 franchises
Win % this season: 95% CI with clustered errors [-968, 3,706]
Win % last season: 95% CI with clustered errors [1,666, 7,484]
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Bruesch-Pagan Test of constant variance. (LM = 106.0, p = 7.9 * 10^-7). Clustered by Franchise, 32 clusters. Clustered 95% intervals: -968 to 3706, this season. 1666 to 7484, last season. VIFs: 1.4 for each win % variable, 1698.6 for log metro population.

Table A3: The five most influential team seasons in the main model

team-season	attendance	leverage	std resid	Cook D
Los Angeles Rams 2016	83,165	0.290	4.19	0.156
Los Angeles Rams 2017	63,392	0.286	-3.46	0.104
Washington Redskins 2018	61,028	0.129	-5.27	0.090
Minnesota Vikings 2015	52,430	0.133	-3.73	0.047
Minnesota Vikings 2014	52,238	0.133	-3.66	0.045

Cook's D above 4/n (0.0125): 21 of 320 team-seasons

Without Los Angeles Rams 2016:
  Win % this season: 1,369 -> 1,939 (moved 0.57 of its standard error)
  Win % last season: 4,575 -> 4,739 (moved 0.16 of its standard error)

21 of 320 team seasons have a Cook's distance above 4/n = 0.0125. Without Los Angeles Rams 2016, the coefficient on this season's win % moves from 1,369 to 1,939 (0.57 of its standard error) and on last season's from 4,575 to 4,739 (0.16 of its standard error).